

in i5 that's lacking in i3. This technology enables the processor to run above its base operating frequency via dynamic control of the processor's clock rate. It's activated when the OS requests the highest performance state of the processor. Most other features are almost the same. Core i7 series, however, offers four core processors, each with HT technology making them seem like octa-core processors.

Again, there are generations for each of these series. The most recent generation, launched in June, is 4th generation. As we write this, 4th generation Core i7 and Core i5 mobile processors have been introduced and updated on Intel's website. The easiest way to find out the generation of an Intel Core mobile processor is to see the first number of the four-digit processor number. For example, both i5-3550 and i3-3240 are 3rd generation processors while i5-2500S and i3-2130 are 2nd generation processors. In case an Intel Core processor number is just three digits, it belongs to the first generation.

AMD processors, on the other hand, are not as easy to identify. The company's technical documentation page provides definitions for each alpha-numeric character of the OPN (Ordering Part Numbers – the laser markings on AMD processors that identify processors and their specs) on the power and thermal data sheet for each processor family. Follow these links for details of the model numbers: <http://dgit.in/17ZED7M> and <http://dgit.in/19TPOzu>

Specifications

Below are some important specs in a processor you should analyse when deciding on a laptop. We'll break down and explain each specification:

1. **Clock Speed:** The “more is better” rule applies here but you must remember that a higher clock-speed translates to higher power consumption and shorter battery back-up time. Wisdom lies in understanding the major use of your laptop. If you're not going to do computation intensive multimedia jobs, don't be lured by big numbers. For most regular work such as document editing, web browsing or programming, the computation load remains low whereas multimedia jobs such as graphics designing, video editing, 3D modelling and animation require more CPU power.

Note, however, that while the clock speed-to-performance ratio is stable within the same family of processors, the amount of work done per clock cycle is different for each processor architecture and family. Many more